

08 — Technical Memoranda (Traffic, Sewer, Water, Stormwater, Fire, Schools, Environmental)

Project: The Cottages at Arcade Springs — R-1 → R-2 rezoning, 4541/4539/4537/4535 Arcade Rd SW, Lilburn, GA (PINs 6123 033, 015, 014, 162; Land Lot 123, 6th District; 9.44 ac deeded / 9.58 ac GIS) **Prepared:** 2026-08-29 · **Revised 2026-09-03** to the external-fact re-verification of that date · **Re-issued 2026-09-03** on the corrected site plan, at which the lot count fell from 43 to **41** (see both revision notes below) **Status:** DRAFT — NOT SEALED. To be superseded by the sealed work of a Georgia PE (memos A–E) and by GCPS / GA EPD correspondence (memos F–G). Nothing here is a design; every memo is a feasibility screen sized to the rezoning question "can this site be served?"

Revision note — what changed on 2026-09-03 and why. Four positions in the 2026-08-29 issue of this document were wrong on the primary sources and have been rewritten, not patched:

Memo	Was	Is
A	"at 12.9 peak-hour trips no formal study is expected"	A Level-1 TIS is required. Gwinnett DOT: "A TIS must be submitted concurrently with all zoning applications"; the only residential exemption is seven lots or units. The 2017 GDOT count is also ineligible under the county's own two-year currency rule
A	LUC 251 rates quoted from two "11th-edition" compilations	Both compilations carry pre-11th-edition land-use names. The rate is UNRESOLVED ; a range is carried and the conclusion is shown to be insensitive to it
B	Phase 2 "primary" route = in-tract HOA lift station or grinder pumps	Foreclosed by policy. Gwinnett DWR's <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018) allows private pump stations, force mains and gravity sewers "only ... for commercial properties under single ownership"; WSR-24 §1.3.1(A) allows a county station only where gravity is "more than 5,000 feet down gradient" against an actual 178 ft; and Gwinnett has no grinder-pump or low-pressure-sewer standard. The 178-ft off-site gravity tie is the only viable Phase-2 route. Phase 1 needs no extension at all
E	"43 units (> 30) → two access roads unless sprinklered; NFPA 13D is the mitigation"	Ga. Comp. R. & Regs. 120-3-3-.04 raises the Appendix D107.1 trigger from 30 to 120 dwelling units. At 41 units the two-access requirement is not engaged . NFPA 13D sprinklers are a voluntary offer , not a required mitigation. The live fire-access item is the 1,754-ft dead end under Table D103.4

Two further corrections run through the whole document: the fire-code citation form (the operative Georgia modifications are issued by the Safety Fire Commissioner at **Ga. Comp. R. & Regs. 120-3-3-.04**, not as a DCA amendment packet), and the impervious area, which the 2026-08-29 issue understated by about 24,000 SF (160,167 SF reported against the 184,148 SF the corrected plan actually draws — see D.2).

Revision note — the corrected site plan (second issue of 2026-09-03), and what it does to these memos. After that revision was written, the site-plan generator was corrected against the geometry audit in audit-2026-09-03/site-geometry.md. Two of its findings cost lots. Every lot is now screened against **all three** digitised stream reaches rather than only the on-site one (SG-04), which put the southwest slot at u 1,230–1,280 inside a buffer and turned it into an unlotted open-space tract; and the stormwater basins, which had been sized on an assumed 4.4 disturbed acres against a real 7.5 (SG-01), had to be enlarged into ground that had held a lot. **The yield is 41, not 43.** Every per-unit and per-acre input in these memos moves with it, and so does everything computed from the basin sizes and the impervious area:

Quantity	2026-08-29 / first 09-03 issue (43 lots)	This issue (41 lots — data/layout.json)	Memo
Dwelling units	43	41 (14 southwest, 27 northeast)	all
Phase split	24 + 19 at u = 1,000 ft	24 + 17 at the drawn phase line u = 980 ft	B
PM peak-hour trips at 0.30/DU	12.9	12.3	A
Daily trips (mid rate)	185	177	A
Average daily sewer flow, DWR table	11,170 gpd	10,670 gpd	B
Average-day water demand	10,750 gpd	10,250 gpd	C
Impervious area	188,483 SF / 45.2 %	184,148 SF / 44.1 %	D
Disturbed area	327,280 SF / 7.513 ac	328,484 SF / 7.541 ac	D
Detention required / provided	75,133 cf / 44,218 cf (59 %)	75,410 cf / 77,863 cf (103.3 %)	D
Water-quality volume WQv	19,049 cf	18,659 cf	D
HOPA units needing a 55+ resident	35 of 43	33 of 41	F
Voluntary NFPA 13D offer	≈ \$129,000	≈ \$123,000	E
Off-site sewer, per Phase-2 lot	\$8,800–\$14,400 (19 lots)	\$9,900–\$16,000 (17 lots)	B

Three of these cut against the application and are stated plainly rather than buried. (i) The **off-site sewer cost per Phase-2 lot rises by about 12 %**, because the same ≈ 478-ft extension is now spread over 17 lots instead of 19. (ii) **41 lots is one above the**

bottom of the 40–47 design range of FACTS.md §4, so there is very little room left for a further lot loss at the boundary survey or the stream delineation before the program has to be re-thought. (iii) The corrected entrance geometry leaves **2.0 ft of the southwest curb return inside the 20-ft perimeter buffer band** (about 0.1 SF of pavement, at the right-of-way corner only, leaving a residual undisturbed width of 18'-0" there), so a buffer reduction is now requested in the letter of intent rather than disclaimed — Memo A.4 explains why the entrance cannot simply be shifted northeast. Working the other way, the enlarged basins now **over-provide** detention instead of falling 41 % short, and the impervious percentage falls slightly.

Program basis (data/layout.json, data/plans.json; **the site plan sheet governs over any number repeated here**): **N = 41** one-story detached cottage homes on fee-simple 50'-0" × 100'-0" lots (5,000 SF each — 205,000 SF of lots; acceptable design range 40–47, cap 47 by voluntary condition 2), 55+ (HOPA) as a voluntary condition, one private lane (22-ft pavement in a 35.9–46.1-ft tract, widening to 60 ft through the amenity block, **1,754 ft** long measured along the travelled way from the Arcado Rd right-of-way, dead-end with three hammerhead turnarounds at stations 594.5 / 1,164.5 / 1,744.5 ft), one driveway to Arcado Rd, a ≈ 2,400 SF clubhouse, two dry detention / water-quality ponds, common open space **141,089 SF (3.24 ac = 33.8 % of the GIS area, 34.3 % of 9.44 ac)**, of which 13,960 SF is built or paved, impervious **184,148 SF (44.1 % of the GIS area)**. Gross density **4.34 du/ac** on 9.44 ac (4.28 on the 9.58-ac GIS area). Two phases at the u = 980-ft phase line (**24 + 17 lots**; voluntary condition 15 describes the same line as "u ≈ 1,000 ft"). Site-local coordinates (u along the strip from Arcado Rd, v across; feet) are those of FACTS.md §1.

Editions cited: ITE *Trip Generation Manual*, 11th ed. (2021); Gwinnett DOT *Traffic Impact Study Guidelines* (2023); Gwinnett DOT *Criteria and Guidelines for Left Turn Lanes* (2007-06-20); Gwinnett County Unified Development Ordinance §900-40.4 Table 900.2 and §900-40.5 Table 900.3 (Ord. No. UDOA2014-00001, 7-22-14); Gwinnett DWR *Water Main and Sanitary Sewer Design and Construction Standards and Specifications* (April 2016); Gwinnett DWR *Wastewater Flow Estimation Guidelines* (Rev. 10/2022); Gwinnett DWR *Sewer Capacity Certification Request* (Rev. 07/2023); Gwinnett DWR *Developer Pump Station Standards* (January 2014, posted by Planning & Development as WSR-24); **Gwinnett DWR Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. September 2018)**; Gwinnett County Stormwater Management Manual (GCSWMM) / Georgia Stormwater Management Manual (GSMM 2016); **IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04, "State Minimum Fire Safety Standards with Modifications" (Safety Fire Commissioner; O.C.G.A. Title 25 Ch. 2), Appendices B and D**; NFPA 13D; Lilburn Zoning Ordinance 2023-603; Lilburn Development Regulations Appendix B Article 5 §5.9; Lilburn Code Ch. 109 Art. V and Art. VII; O.C.G.A. §§12-7-6, 25-2-12(e)(4).

A caution on the fire-code citation. The copy of Ga. Comp. R. & Regs. 120-3-3 posted at oci.georgia.gov is stale — it still recites the 2018 edition of the fire code. The Secretary of State's compiled version at rules.sos.ga.gov reflects the 2024 edition. Cite the SOS compilation and confirm the effective text with the Gwinnett County Fire Marshal [VERIFY].

Memo A — Traffic

Question. How many trips does the cottage program add to Arcado Rd, **what study does Gwinnett DOT require**, is a left-turn lane warranted, and does the access geometry meet the county's spacing and sight-distance standards?

A.1 The threshold question is settled: a study is required

Gwinnett DOT, *Traffic Impact Study Guidelines* (2023), on applicability:

"A TIS must be submitted concurrently with all zoning applications and land disturbance development."

and, in the exemptions:

"Exemptions — A TIS shall not be required under the following conditions: • Single-family detached and townhome residential development consisting of no more than seven lots or units ..."

At **41 lots** this project is not exempt (41 > 7). The 2026-08-29 issue of this memo already reached that conclusion; FACTS.md §3's "below 21 peak-hour trips → no formal TIS expected" and every document that repeated it are **wrong** and are being conformed. The trip count does not decide *whether* a study is required — it decides *which level*.

The guidelines are written for development "within unincorporated Gwinnett County" under Gwinnett UDO §270-20.8. The City of Lilburn is not bound by them for a city zoning case, but (a) Gwinnett DOT owns Arcado Rd and will apply them at the driveway-permit stage regardless, and (b) staff and neighbours will read them as the standard. **Ask the Planning Director at the pre-application conference whether the City adopts them for a city case and at what level** [VERIFY]. The applicant is preparing a Level-1 study either way, and offers it as voluntary condition 17.

A.2 Inputs

Input	Value	Source
Units	41 (cap 47 by voluntary condition 2)	data/layout.json metrics.lots
ITE LUC 251 rate — UNRESOLVED	AM 0.20–0.34, PM 0.26–0.39, daily 3.7–4.31 trips/DU	see A.3
LUC 210 <i>Single-Family Detached Housing</i> , 11th ed.	daily 9.43, AM 0.70, PM 0.94	Palm Beach County Engineering trip-rate sheet
LUC 220 <i>Multifamily Low-Rise</i> , 11th ed.	daily 6.74, AM 0.40, PM 0.51	same
LUC 822 <i>Strip Retail Plaza (< 40 ksf)</i> , 11th ed.	daily 54.45, PM 6.59 per 1,000 SF	same
As-zoned R-1 yield	4 du/ac × 9.44 ac = 37.8 → 37 lots	Lilburn Zoning Ordinance 2023-603 Table 4.1 (R-1 max 4 du/ac)
2025 MU program (RZ-2025-01)	104 condominiums + 24,400 SF commercial	status/rz-2025-01/4814_development-summary-report.pdf
Arcado Rd	county minor collector, 2 lanes, posted 35 mph	Gwinnett GIS; FACTS §2b
Arcado Rd volume	no eligible count exists — the 2017 GDOT count (station 135-6689, 14,800 AADT) is nine years old	see A.6
TIS study levels	0–20 peak-hour site trips = Level 1 ; 21–249 = Level 2; 250–499 = Level 3; > 500 or DRI = Level 4	Gwinnett DOT TIS Guidelines (2023), Tables 1 and 2
Left-turn-lane warrant	Table 1 (Residential Developments), 2-lane route, posted 30 to 35 mph , ADT ≥ 6,000 → 75 Lots	Gwinnett DOT <i>Criteria and Guidelines for Left Turn Lanes</i> (2007-06-20), Table 1
Driveway spacing	posted 35 mph → 244 ft , "measured from centerline to centerline of driveways, public roads or side streets"	Gwinnett UDO §900-40.5, Table 900.3
Maximum driveways	"one for each 400 feet of property frontage, or fraction thereof"	Gwinnett UDO §900-40.5(C)
Intersection sight distance	35 mph, 2-lane, SDL = SDR = 390 ft , on the AASHTO Case B1 basis (the distance travelled at the arterial speed during 7.5 seconds)	Gwinnett UDO §900-40.4, Table 900.2
Provided geometry	entrance centred at v = –190; 251 ft from the Arcadia PI centreline measured along the Arcado Rd centreline (251.2 ft along the centreline, 250.7 ft chord); frontage 237.4 ft chord / 237.6 ft along the R/W; one driveway	data/layout.json lane.entrance, metrics

A.3 The LUC 251 rate is UNRESOLVED — and it does not matter to the conclusion

The ITE *Trip Generation Manual*, 11th ed. is a paywalled publication; no primary source was reachable on 2026-09-03. Four secondary compilations were retrieved and reconciled:

Compilation	Land-use name used	AM	PM	Daily
Palm Beach County Engineering trip-rate sheet	"Senior Adult Housing- Detached "	0.24	0.30	4.31
Phoenix, Oregon SDC table (labelled "ITE 11th Edition")	"Senior Adult Housing- Detached "	—	0.30	—
City of Corpus Christi "Common Trip Generation Rates – 11th Edition"	"Senior Adult Housing- Detached "	0.34	0.39	—
Falls Church VA <i>Railroad Cottages</i> study (states ITE 9th ed.)	"Senior Adult Housing Det."	0.20	0.30	3.70

None of these is an 11th-edition source. DelDOT's official *Trip Generation 11th Edition Updates* briefing records that the land use was renamed and re-scoped in the 11th edition: "New Land Uses: ... **Senior Adult Housing-Single Family (LUC 251) – Combination of detached and senior adult housing-attached**; Senior Adult Housing-Multifamily (LUC 252)." Every table above still uses the pre-11th name "Senior Adult Housing-**Detached**," and the Corpus Christi sheet's neighbouring rows ("220 Apartment," "230 Residential Condominium/Townhouse") are 9th-edition land-use names. The 2026-08-29 issue of this memo described two of these as "the two published 11th-ed. compilations"; that description was wrong and is withdrawn.

What would settle it: the LUC 251 data plots in the ITE *Trip Generation Manual*, 11th ed. itself, or an ITETripGen web-app query, read by the traffic engineer preparing the Level-1 study. That is a one-page task for anyone with an ITE subscription and it should be done before filing. Until then this memo carries the **range**, and states the rate source as UNRESOLVED wherever a single number is quoted elsewhere in the package (the package's headline figure, **12.3 PM peak-hour trips**, is 41×0.30 and sits in the middle of the range).

A.4 Results

Trips = $N \times \text{rate}$.

Scenario	Units	Daily	AM peak	PM peak
Proposed, low rate pair (0.20 AM / 0.30 PM / 3.70 daily)	41	152	8.2	12.3
Proposed, mid (0.24 / 0.30 / 4.31)	41	177	9.8	12.3
Proposed, high (0.34 / 0.39)	41	—	13.9	16.0
Proposed at the 47-unit cap, high rate pair	47	203	16.0	18.3
As-zoned R-1 by right, LUC 210	37	349	25.9	34.8
2025 MU program, raw (no internal capture or pass-by)	104 DU + 24.4 ksf	≈ 2,030	—	≈ 214 (104 × 0.51 = 53, plus 24.4 × 6.59 = 161)

- **Study level: Level 1, at every rate pair and at the 47-unit cap.** The highest figure anywhere in the table for the proposed program is **18.3** PM peak-hour trips, inside the 0–20 band. The conclusion does not turn on the unresolved rate — that is the point to make to staff.
- Against the as-zoned R-1 subdivision the cottage program is about **–49 %** daily (177 against 349) and **–65 %** in the PM peak (12.3 against 34.8); against the 2025 mixed-use filing it is about **–94 %** in the PM peak (12.3 against ≈ 214). The two-lot loss improves each of these comparisons slightly; none of them turns on it.
- **Left-turn lane: not warranted.** 41 lots (and 47 at the cap) are below the **75-lot** threshold for a 2-lane route posted at 30–35 mph carrying ADT ≥ 6,000. A left-turn lane would also be physically difficult: the Arcado Rd centreline is only about 28–30 ft from the front property line.
- **Driveway spacing: appears consistent.** 251 ft provided against 244 ft required (UDO Table 900.3) — a margin of about 7 ft, which is why the entrance cannot be shifted northeast to cure the buffer clip at the southwest curb return. That clip is now 2.0 ft measured along the right-of-way — a residual buffer width of 18'-0" instead of 20'-0", over about 0.1 SF of pavement — after the southwest curb return was reduced from 25'-0" to 15'-0"; it is the buffer reduction requested in the Letter of Intent §5 and carried in voluntary condition 4.
- **Maximum driveways: one.** At ≈ 237 ft of frontage, §900-40.5(C) permits one driveway. The single-entrance plan is not a concession, it is the county standard.
- **Sight distance: 390 ft each way**, to be certified by a Georgia PE on the curved frontage; clearing within the 10-ft landscape strip and the right-of-way may be required [VERIFY by field survey].

A.5 Scope of the Level-1 study (the deliverable)

Table 1 of the guidelines marks these elements for a Level 1 study, and this memo already contains the substance of every one of them:

1. Location description and site access description.
2. Existing and proposed land use; zoning request.
3. Trip-generation estimate (A.3–A.4), with the ITE land-use code, edition and rate stated.
4. Access-management review against UDO §900-40.
5. **Adjacent access spacing, upstream and downstream** — Arcadia Pl at 251 ft; the next driveways on both sides of Arcado Rd to be measured centreline-to-centreline in the field.
6. **Intersection sight distance** — 390 ft, PE certificate.
7. Connectivity and circulation review — one entrance, 1,754-ft private lane, three hammerhead turnarounds, no through connection possible.
8. Functional classification and posted speed — minor collector, 35 mph.
9. Future identified projects on the corridor (Killian Hill Rd SPLOST project F-1262 status [VERIFY]).

Operations analysis, turn-lane warrant analysis and signal-warrant analysis are **Level 2** elements and are not required here (Table 2).

A.6 The traffic count must be new

Gwinnett DOT, *Criteria and Guidelines for Left Turn Lanes*:

*"GCDOT traffic counts shall be used **provided they are less than two years old** ... If GCDOT traffic counts are not available, or they are greater than two years old, Georgia DOT (GDOT) counts may be used **provided they are also less than two years old**. If counts are not available from either source, and/or both sources are greater than two years old, **the developer shall be responsible for obtaining new, 24-hour counts, from an approved contractor, as directed by GCDOT.**"*

The 2017 GDOT count (station 135-6689, 14,800 AADT, 2.2 % trucks) is nine years old and does not satisfy the rule. It is retained in this package only as historical context and must not be used as the study's base volume. **Budget a 24-hour tube count on Arcado Rd at the site frontage, by a GCDOT-approved contractor** — a small cost, and one that also refreshes the ADT $\geq 6,000$ input to the left-turn-lane test. Order it before the study is written.

A.7 Next steps

1. Traffic engineer: read LUC 251 (and 210 for the by-right comparison) from the ITE 11th-ed. manual; close A.3.
2. Commission the 24-hour count; obtain the sight-distance certificate.
3. Gwinnett DOT pre-submittal contact — the guidelines allow scope to be adjusted or waived at a pre-submittal meeting.
4. Ask Lilburn Planning, in writing, what study the **City** wants for a city zoning case.

Memo B — Sanitary Sewer

Question. Which lots drain by gravity to the existing "Arcado Road Townhomes Sewer Outfall" main on the property, why the rear pocket cannot, **what routes Gwinnett DWR's policies actually permit**, and what DWR needs before the zoning decision.

B.1 Inputs

Input	Value	Source
Existing 8-in gravity main on site	along the SW edge inside parcel 033 ($v \approx -224$ to -228), $u \approx -42 \rightarrow 272$ (≈ 314 ft on the property), then SW off site	FACTS §2; layout . json sewer.existing_on_site (Gwinnett DWR GIS)
Inverts (GIS attributes, unsurveyed)	943.79 / 940.47 (Arcado) $\rightarrow$ 932.69 / 930.84 ($u -42 \rightarrow 20$) $\rightarrow$ 930.64 / 927.13 at the MH near (u 272, v -224) $\rightarrow$ 925.17 downstream	FACTS §2
Lane ground profile (USGS 3DEP, approximate)	u 279: 946.4 (hump) · u 779: 936.3 (sag) · u 979: 940.4 · u 1,179: 953.5 (ridge) · u 1,479: 931.5 (rear low) · u 1,699: 947.4	layout . json lane.ground_profile (10-ft stations)
Legends at Parkview 8-in main	MH invert 919.58 near (u 1,349, $v -405$); 921.03 $\rightarrow$ 919.58 along u 1,160–1,349	FACTS §2
Phase-2 gravity route	rear low point (u 1,479) $\rightarrow$ lane u 1,300 $\rightarrow$ SW line $\rightarrow$ Legends MH: $\approx$ 478 ft total (179 + 121 + 178), of which the last $\approx$ 178 ft is off-site	layout . json sewer.phase2_alternative_extension
Design assumptions	8-in PVC/DIP; minimum slope 0.40 % ; Manning $n = 0.013$; 5-ft cover to invert at the upstream end (one-story slab houses, ~ 4 -ft laterals)	Gwinnett DWR Standards (2016) §2.3 [VERIFY Gwinnett's tabulated minimum slope for 8-in]
Flow basis	Single-family home 250 gpd/unit ; clubhouse 175 gpd per 1,000 SF; peaking factor 4.0 ("the peaking factor has been changed from 2.5 to 4.0 unless justification can be provided")	Gwinnett DWR <i>Wastewater Flow Estimation Guidelines</i> (Rev. 10/2022)
Private-facility policy	"Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development "	Gwinnett DWR <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018)
County pump-station policy	a station is permitted only where gravity is " more than 5,000 feet down gradient "	Gwinnett DWR <i>Developer Pump Station Standards</i> (Jan 2014, WSR-24) §1.3.1(A)
Grinder pumps / low-pressure sewer	no Gwinnett standard exists — zero occurrences of "grinder," and no low-pressure sewer system provisions, in the <i>Water and Sewer Standards</i> (April 2016, 117 pp.) or in WSR-24	full-text search, 2026-09-03
Comprehensive-plan policy	"Do not further extend sewer in this area" (Suburban-Low, 2026 CPA draft of 6/16/2026, p. 73) — transmitted to DCA 2026-07-13, not adopted as of 2026-09-03	FACTS §2b; audit external-facts.md §3.3

B.2 Formulas

Available slope $S = (\text{invert}_{\text{upstream}} - \text{invert}_{\text{downstream}}) / L$. Full-flow capacity $Q = (1.49/n) \cdot A \cdot R^{(2/3)} \cdot S^{(1/2)}$. AADF = $\Sigma(\text{units} \times \text{unit flow})$; peak = $4.0 \times \text{AADF}$; gpm = gpd $\div$ 1,440.

B.3 Phase 1 (24 lots, $u \leq 980$) — gravity, and no extension of the public sewer

- Route: the lane from the $u = 980$ -ft phase line back to u 272 (708 ft), then 106 ft across the amenity tract to the existing manhole at (271.7, -224): **$L \approx 814$ ft**.
- Upstream: invert at u 980 = $940.4 - 5.0 = \mathbf{935.4}$. From there to the lane sag at u 779 (invert $936.3 - 5.0 = \mathbf{931.3}$) the pipe falls 4.1 ft in 200 ft = **2.05 %**.

- **Controlling reach, sag → manhole:** $L = (779 - 272) + 106 = 613$ ft; $S = (931.3 - 927.13) \div 613 = 4.17 \div 613 = 0.68$ %, against the 0.40 % minimum. **Gravity works.** The two lots the plan lost do not touch this: the phase line and the controlling reach are fixed by ground elevation, not by yield.
- The buildability catch is the **lane hump at $u \approx 279$ (ground 946.4)**: a pipe leaving the sag at the 0.40 % minimum is at $931.3 - 0.004 \times 500 = 929.3$ there, about **17 ft to invert** — a deep-cut item (trench boxes, possible rock in the ARE map unit). Alternative alignment: run the front trunk in the **SW-edge easement corridor** ($v \approx -200$ to -225), where the 3DEP ground is about 939 at u 250 and the cut is roughly **10 ft**. The PE chooses; either way the answer to the comprehensive-plan sewer policy for Phase 1 is that **nothing is extended**. The main is already inside the property line.
- Capacity: an 8-in pipe at 0.40 % carries about 0.77 cfs = **344 gpm** full. The whole project's peak is 29.6 gpm (B.6), under 10 % of pipe capacity. The constraint is the downstream county system, not the on-site pipe — which is exactly what the capacity certification tests.

B.4 Phase 2 (17 lots, the rear pocket) — why gravity to the on-site main fails

- Rear lane low point u 1,479 at ground **931.5**; an invert there with 5 ft of cover is **926.5** — already **0.6 ft below** the 927.13 manhole invert before any slope is applied.
- At the 0.40 % minimum over $L = (1,479 - 272) + 106 = 1,313$ ft, the pipe would arrive at $926.5 - 5.25 = 921.2$, a **5.9-ft deficit** against 927.13; and to get there it would pass beneath the $u \approx 1,179$ ridge (953.5) at roughly **28 ft** of cut. **Not buildable by gravity to the on-site main.** The ridge is the reason the project is phased at all.

B.5 The two rear options, decided

Option 1 — 178-ft off-site gravity tie to the Legends at Parkview main: the only viable route

Item	Value	Basis
Total length	≈ 478 ft (179 ft along the lane + 121 ft across to the SW line + 178 ft off site), of which ≈ 178 ft off-site	layout . json sewer.ph ase2_alternative_extensi on
Fall	$926.5 \rightarrow 919.58 = 6.92$ ft	5-ft cover at the rear low point; Legends MH invert
Slope	$6.92 \div 478 = 1.45$ %	comfortable against the 0.40 % minimum
Depth — the weak point of the drawn alignment	the route as drawn runs back up the lane from the rear low point (931.5) to u 1,300, where existing ground is 943.8–944.7: the pipe is about 20 ft deep there, and about 10.5 ft deep where it reaches the SW property line at ($1,300, -235$), ground 932.7	layout . json lane.ground_profile; dat a/topo-samples . json
Depth, alternative alignment	leave the lane at the rear low point and run southwest through the low pocket (3DEP ground 921.6–929.5 at u 1,350–1,450, v -140 to -235 — the Pond 2 / creek-woods side); cover then stays 5 to 10 ft the whole way. The PE should price both. This memo carries the drawn route's length and slope because the drawing governs, and flags the 20-ft cut rather than hiding it	3DEP profile
Manholes	3 (rear low point; at the SW property line; connection at the Legends MH)	DWR Standards (2016) spacing
Capacity used	Phase-2 peak 11.8 gpm into an 8-in main	B.6

What it needs.

1. **A recorded permanent sanitary sewer easement** across one Village Green Court lot, or along the stream corridor, in the Legends at Parkview subdivision. This is a private negotiation with a private owner (and possibly the Legends HOA); the City cannot grant it and the applicant cannot promise it. Pull the current registered agent and officers of Legends at Parkview Homeowners Association, Inc. from ecorp.sos.ga.gov before approaching anyone [VERIFY — the "Liberty Community Management" attribution in docs/06 is unconfirmed].
2. **A buffer crossing that the stream ordinance permits.** The route crosses the stream corridor. The Site Development Plan Review Checklist allows utility encroachment into a buffer up to 50 ft of width measured perpendicular; the MNGWPD model ordinance that Lilburn adopted keeps sewer easements at least 25 ft from the top of bank where practicable. Design the crossing **perpendicular, ≤ 50 ft wide**, restored under the 2019 Lilburn Riparian Buffer Restoration Guidelines [VERIFY the Lilburn Code Ch. 109 Art. VII section numbers by hand from Municode — automated retrieval returns HTTP 403].
3. **A Gwinnett DWR sewer capacity certification** for the Legends basin (B.7).
4. **A staff reading of the comprehensive-plan sewer policy** (B.8).

Order-of-magnitude cost — planning estimate only, unit rates are assumptions to be replaced by contractor quotes [VERIFY]:

Item	Quantity	Assumed unit rate	Amount
8-in PVC gravity sewer, 5–20 ft cut, in a wooded easement	478 LF	\$150–\$225 / LF	\$71,700–\$107,600
Precast manholes, 4-ft dia., with drop/doghouse connection at the Legends MH	3 ea	\$7,000–\$10,000 ea	\$21,000–\$30,000
Rock excavation allowance (ARE map unit, gneiss at 22–40 in)	200 CY	\$150 / CY	\$30,000
Stream/buffer crossing, erosion control and restoration	lump	—	\$10,000–\$20,000
Off-site easement acquisition from one private owner	1	negotiated; no market source	\$15,000–\$50,000
Survey, PE design, DWR certification, inspection, as-builts	lump	—	\$20,000–\$35,000
Order-of-magnitude total			≈ \$168,000–\$273,000

That is roughly **\$9,900 to \$16,000 per Phase-2 lot** (17 lots) — up about 12 % per lot from the \$8,800–\$14,400 the 2026-08-29 issue carried, because the same extension is now spread over 17 Phase-2 lots instead of 19. It is material, it is worse than it was, and it is the honest number to carry into the pro forma. The three items that could move it are rock, the easement price and the 20-ft cut at the lane crossing, and none of them can be closed from a desk.

Option 2 — in-tract pump station, force main or grinder pumps: FORECLOSED BY POLICY

The 2026-08-29 issue of this memo, following FACTS.md §2b, presented this as the *primary* Phase-2 route. It is not available. Three independent bars:

(a) A private (HOA-owned) station or force main. Gwinnett DWR, *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions* (rev. September 2018) — the policy variant whose cover carries a "Subdivisions" check-box, and therefore the one that governs a fee-simple lot subdivision:

"PRIVATE FACILITIES: Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development. Gwinnett County is not responsible for operation and maintenance of these facilities."

and, immediately above it:

"OWNERSHIP OF MAINS AND APPURTENANCES: Absolute title and ownership of all water and sanitary sewer mains within the private development shall be conveyed to Gwinnett County WSA at the time of acceptance by the County."

A 41-lot fee-simple subdivision is not "commercial property under single ownership." (The 2026-08-29 memo's judgment cited the **Apartments** variant of the same policy; the Subdivisions variant is the applicable one and it carries the identical clause.)

(b) A county-owned station. Gwinnett DWR, *Developer Pump Station Standards* (January 2014, WSR-24) §1.3.1(A), in full:

"Approval for the installation of a new pump station will only be granted by the GCDWR's AMIS Division Director through consultation with the GCDP&D. Generally, pump stations will only be permitted when gravity sewer is unavailable to the property. Unavailable shall generally be interpreted to mean more than 5,000 feet down gradient, but this distance can be increased or decreased by GCDWR Division Directors based upon actual field conditions and the size of the project involved."

Gravity is **178 ft** away — two orders of magnitude inside the 5,000-ft test. The final clause is the only opening, it cuts both ways (the Director may decrease the distance as well as increase it), and it is exercised case by case. The January 2014 document is the current edition: the copy DWR posts as "current" is byte-identical (MD5 0fe478d40303252f12497f6996b7ee49).

(c) Grinder pumps or a low-pressure sewer system. There is **no Gwinnett design standard for either**. A full-text search of the *Water and Sewer Standards* (April 2016) returns zero occurrences of "grinder" and no low-pressure sewer system provisions (the only "low pressure" hits are the §5.7 low-pressure **air testing** procedure); WSR-24 returns zero occurrences of "grinder." A system with no county standard could only be private — which clause (a) forecloses.

Conclusion. Option 2 is withdrawn from the application. The Letter of Intent §7, the Development Summary §5.2, voluntary condition 16 and the site-plan general notes have been conformed. **Two artefacts of the withdrawn option are still in the drawing data and must come out before filing:** data/layout.json still carries sewer.phase2_primary describing "in-tract lift station / grinder pumps ... force main to the Phase 1 gravity main" as the primary Phase-2 route, and a lift_station_symbol polygon at (u 1,481–1,489, v –146.6 to –138.6). The same file's own screening table already states the opposite — that no pumping solution is available and that the 178-ft gravity tie is the Phase-2 route — so the file contradicts itself. Delete both, and delete the LS symbol from the site plan [VERIFY on the next regeneration].

Option 3 — Phase 1 alone (24 lots)

If the easement or the capacity certification cannot be obtained, Phase 1 stands by itself: 24 lots, the amenity block, Pond 1 and the entrance, all on gravity to the existing on-site main, at about 2.5 du/ac. Voluntary conditions 15 and 16 make that the

enforceable default, which is why the applicant can ask for R-2 on the whole tract without asking the City to take the Phase-2 risk.

B.6 Estimated flows

Basis	AADF	Peak (× 4.0)
DWR table (governs the certification): 41 × 250 gpd + 2,400 SF clubhouse × 175 gpd per 1,000 SF (420 gpd)	10,670 gpd = 7.41 gpm	42,680 gpd = 29.6 gpm
Phase 1: 24 lots + clubhouse (24 × 250 + 420)	6,420 gpd (4.5 gpm)	25,680 gpd (17.8 gpm)
Phase 2: 17 lots (17 × 250)	4,250 gpd (2.95 gpm)	17,000 gpd (11.8 gpm)
Cross-check, population basis: 41 × 2.2 persons × 70 gpd	6,314 gpd (4.4 gpm)	25,256 gpd (17.5 gpm)

The DWR table figures govern; the population-basis row is shown only to bracket them. The 300 gpd/unit figure used in earlier drafts of docs/02 and docs/10 is retired. For comparison, the 2025 mixed-use filing (104 condominiums plus 24,400 SF of commercial) would have requested roughly two to three times this average daily flow [VERIFY against 4819_sewer-capacity-request.pdf, an image scan].

B.7 The DWR capacity certification — and the exact question to ask

The request goes on the *Sewer Capacity Certification Request*, **Rev. 07/2023**, using its **"Pre-Rezoning"** request type. The form must be signed and sealed by a design professional (this is not owner-preparable), is submitted to **DWRCapacityCertification@GwinnettCounty.com**, and asks for the target zoning-submittal and zoning-approval dates — DWR expects to be engaged before the zoning decision. Turnaround: **"PLEASE ALLOW A MINIMUM OF 10 BUSINESS DAYS FOR PROCESSING,"** and **"AN ADDITIONAL 20 BUSINESS DAYS"** if analysis or flow monitoring is required — so **2 to 6 weeks** after a complete package. No fee is stated on the form or in the flow-estimation guidance [VERIFY with DWR].

Attach: a GIS map highlighting the site, surrounding utilities and the **tie-in manhole facility IDs** — the u ≈ 272 manhole (Arcado Road Townhomes outfall, invert 927.13) for Phase 1 and the Legends at Parkview manhole (invert 919.58) for Phase 2; the flow calculations of B.6; and the phasing.

Submit this question, in these words, with the form:

*"The project is a 41-lot fee-simple single-family detached subdivision in the City of Lilburn (PINs R6123 033, 015, 014, 162), phased. Phase 1 (24 lots, AADF 6,420 gpd) connects by gravity to the existing Arcado Road Townhomes outfall manhole on the property, invert 927.13, and requires no extension. Phase 2 (17 lots, AADF 4,250 gpd, peak 11.8 gpm) cannot reach that manhole by gravity because of an intervening ridge. We understand from the Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. 9/2018) that private pump stations, private force mains and private gravity sewers are allowed only for commercial properties under single ownership, and from Developer Pump Station Standards §1.3.1(A) that a county pump station is permitted only where gravity is unavailable — generally more than 5,000 feet down gradient — whereas gravity here is 178 feet away. We therefore propose the only route we believe the policies permit: an 8-inch gravity extension of approximately 478 feet, of which approximately 178 feet is off site, from the rear of the property southwest to the existing Legends at Parkview 8-inch main at invert 919.58, at approximately 1.45 percent, within an easement to be recorded. **Please confirm (1) that DWR concurs that no pumping solution — county-owned station, HOA-owned station, force main, grinder pump or low-pressure sewer — is available to this project under those policies; (2) whether downstream capacity exists in the Legends at Parkview basin at the stated flows for Phase 2 and in the Arcado Road Townhomes outfall for Phase 1; (3) the facility IDs and any conditions DWR would place on the off-site connection, including easement width and manhole type at the tie-in; and (4) whether DWR requires flow monitoring before certification. Target zoning submittal 2026-09-25; target zoning approval 2026-11-09."***

Then carry the certification number into the Lilburn application ("Water, sewer and fire review approved by Gwinnett County" on the Site Development Plan Review Checklist) and revise it at the land-development-permit stage if the unit count changes.

B.8 The comprehensive-plan sewer policy — how to ask

The Suburban-Low policy in the 2026 amendment reads "Do not further extend sewer in this area." Two facts should be put to staff together with the question, in writing:

- **Phase 1 extends nothing.** It connects to a main that already crosses the property.
- **Phase 2 is a 178-ft connection between two existing county mains in the same drainage basin.** It adds no service area beyond this property and opens no unsewered land.

The question for the Planning Director is therefore narrow: *does the City read "do not further extend sewer in this area" as reaching a 178-foot in-basin connection between two existing mains, on a property that is already crossed by a public sewer?* Ask it before filing, and ask the second question with it: **which Future Development class do these four parcels fall in** — Suburban-Low (p. 73, "do not further extend sewer") or Suburban-Medium (p. 71, "coordinate the expansion of sewer infrastructure to fill gaps")? The

Future Development Map (p. 52) is a raster without parcel labels and the answer cannot be read off the document. Both classes rate Single-Family Cottages and Single-Family "Appropriate," so the program survives either way; only the sewer policy differs.

B.9 First spend, before anything else

A **sewer-invert survey** of the $u \approx 272$ manhole and the Legends manhole. Every number in this memo rests on GIS attributes that were never surveyed, and a 1-ft error moves the Phase-1 slope from 0.68 % to about 0.5 % and changes the Phase-2 arithmetic. Add rock probes along the trunk alignment (ARE map unit, gneiss at 22–40 in).

Appears consistent with Gwinnett DWR Standards (2016) for Phase 1. Phase 2 is feasible **only** as the Option 1 gravity tie, and only with the easement and the capacity certification.

Memo C — Water

Question. Domestic demand, fire flow, hydrant layout and the dead-end-main issue.

C.1 Inputs

Input	Value	Source
Existing mains	DIP/ACP in Arcado Rd, 47–107 ft from the front line; nearest hydrant ≈ 61 ft from the front corner ($u -45$, $v +41$)	FACTS §2 (Gwinnett DWR GIS); sizes VERIFY
Unit demand	250 gpd/DU (the DWR wastewater unit flow used as the domestic proxy; HOA-irrigated yards add seasonal demand)	DWR Flow Guidelines (Rev. 10/2022) [VERIFY with DWR's water-model request]
Peaking	max day $2.0 \times$ average; peak hour $4.0 \times$ average	AWWA M22 (2014) practice [VERIFY against DWR model criteria]
Fire flow, one- and two-family dwellings $\leq 3,600$ SF	1,000 gpm for 1 hour; 50 % reduction, not below 500 gpm, where sprinklered per §903.3.1.3 (NFPA 13D)	IFC 2024 (Ga. Comp. R. & Regs. 120-3-3-.04) Appendix B §B105.1 and Table B105.1(1) [VERIFY that Gwinnett's fire-prevention ordinance adopts Appendix B]
Hydrant spacing, residential	"generally located every 400 feet ... spaced from a minimum of 350 feet to an absolute maximum of 450 feet ... on property lines where possible"; "place a fire hydrant near the end of each main"; in-line valves every 1,000 ft or every third hydrant	Gwinnett DWR Standards (2016) §2.2.15(b), (d); §2.2.16(b)
Main length	$\approx$ 1,754 ft single dead-end run in the lane tract	layout.json metrics.lane_length_ft

C.2 Formulas

$Q_{avg} = N \times 250$ gpd; $Q_{maxday} = 2.0 \times Q_{avg}$; $Q_{peakhr} = 4.0 \times Q_{avg} \div 1,440$. Fire flow per Table B105.1(1) with the §B105.1 sprinkler reduction. Hydrants $\approx L \div 400 + 1$.

C.3 Results

- Domestic: **10,250 gpd average** (41×250), 20,500 gpd max day, $\approx$ **28.5 gpm peak hour** ($4.0 \times 10,250 \div 1,440$) — negligible against a collector-road main. The design case is fire flow.
- Fire flow — stated both ways, because the sprinklers are voluntary.** Without sprinklers the requirement is **1,000 gpm at 20 psi residual for 1 hour**. With the NFPA 13D systems offered under voluntary condition 5 in every dwelling, §B105.1 halves it to **500 gpm**. The largest cottage (Plan B "The Laurel", 38'-0" $\times$ 57'-6" body, **1,492 SF conditioned, 2,182 SF under roof** — data/plans.json) is well under the 3,600-SF fire-flow area either way. The clubhouse ($\approx 2,400$ SF, Type V-B) is a separate Table B105.2 check — likely 1,500 gpm unsprinklered, so sprinkler the clubhouse (NFPA 13) to bring it into the same band [VERIFY with the Fire Marshal].
- Hydrants: $\blacksquare 1,754 \div 400 \blacksquare + 1 \rightarrow$ **6 hydrants** (about u 0, 350, 700, 1,050, 1,400 and one at the terminus hammerhead), on lot lines, plus the existing Arcado Rd hydrant at the entrance; 2 in-line valves. Where a hydrant sits on the fire apparatus access road, Appendix D §D103.1 asks for 26 ft of road width — place hydrants at the **hammerhead legs, pocket-green frontages and guest-parking bays**, where the lane locally widens (Memo E).
- Dead-end main:** there is no second frontage, so the main cannot be looped. (i) Use an **8-in main**, not 6-in, so 500–1,000 gpm arrives at the terminus with ≥ 20 psi residual — a 6-in, 1,754-ft dead end is the marginal case; the hydraulic check is DWR's flow test at the Arcado hydrant [static and residual pressures: VERIFY]. (ii) Water age: an 8-in main of this length holds about 4,580 gallons ($\pi \div 4 \times 0.667^2 \times 1,754 \times 7.48$) against about 10,250 gpd of use (turnover under one day), but DWR may still require an **automatic flushing device** or a terminal blow-off (DWR Standards (2016) Detail W4, "Typical Dead End Street Termination"). (iii) Sample taps, not hydrants, for bacteriological testing.

- Meters: individual 5/8-in meters at each lot in the lane tract; one irrigation meter for the HOA; backflow prevention per DWR's cross-connection policy.

Appears consistent with Gwinnett DWR Standards (2016) §2.2.15 and IFC 2024 (Ga. Comp. R. & Regs. 120-3-3-.04) Appendix B — **pending DWR's flow test and the Fire Marshal's concurrence (VERIFY).**

Memo D — Stormwater

Question. GCSWMM approach, impervious area, water-quality and runoff-reduction volumes, detention volumes and pond targets, and the stream-buffer and floodplain statements.

D.1 Inputs

Input	Value	Source
Regulatory framework	The City reviews stormwater under City Code and the Gwinnett County Stormwater Management Manual (GCSWMM) ; two certified stormwater reports at LDP; BMP maintenance agreement plus as-built volume certification	Lilburn Site Development Plan Review Checklist
Site area A	417,174 SF = 9.577 ac GIS (9.44 ac deeded)	layout . json
Impervious	184,148 SF = 4.23 ac = 44.1 % of the GIS area (44.8 % of 9.44 ac deeded)	layout . json metrics.impervious_sf — summed from the drawn footprints; see D.2
Disturbed area	328,484 SF = 7.541 ac (raster site area 418,228 SF less 73,744 SF of 20-ft perimeter buffer bands and 16,000 SF of net creek-woods tract outside those bands = 89,744 SF preserved)	layout . json stormwater.disturbed_area, 2-ft raster; supersedes the "≈ 4.4 ac" assumption the ponds were originally sized on, the "≈ 7.9 ac" upper estimate of 2026-08-29 and the 7.513 ac of the first 09-03 issue
WQv formula (2025 hydrology basis)	WQv = 1.2 · (0.05 + 0.009 · I) ÷ 12 · A (I in %, A in SF → cf)	FACTS §3; RZ-2025-01 hydrology sheet
Detention formula (2025 hydrology basis)	Detention = 10,000 cf per disturbed acre – required runoff-reduction volume (RRv)	same
RRv basis	RRv = 1.0 in · Rv · A ÷ 12, Rv = 0.05 + 0.009 · I	GSMM 2016 Vol. 2 runoff-reduction standard [VERIFY the 2025 sheet used this definition]
Ponds	two dry extended-detention / water-quality basins on the SW side, 6-ft depth, 3:1 side slopes , top of bank 52 × 180 ft (Pond 1) and 64 × 180 ft (Pond 2), in 20,000-SF tracts	layout . json ponds; Sheet C-3.0 (grading and stormwater concept — not yet drawn) will govern once issued
Soils	HSG B Appling/Pacolet-type clays (eroded phases, poor infiltration); ARE (Ashlar, rock at 22–40 in) under both pond sites ; HYB (Helena, HSG D, water table ≈ 24 in) at the front SW corner	FACTS §2b; docs/11 §3.3
Receiving water	Jackson Creek headwaters (GAR030701030315), 2024 303(d) Not Supporting (bacteria, biota; urban runoff)	GA EPD 2024 Integrated 305(b)/303(d) list
Floodplain	FEMA Zone X [FIRM panel and effective date VERIFY]	Gwinnett GIS

D.2 The impervious and disturbed-area corrections, and the arithmetic

Two errors were carried into the 2026-08-29 issue and both have now been rebuilt from the drawing rather than patched.

Impervious. The 2026-08-29 issue reported **160,167 SF (38.4 %)** and data/layout . json then reported 161,212 SF (38.6 %). Both rested on a flat **2,200 SF per lot** allowance hard-coded in the site-plan generator. The generator now sums the drawn footprints, at 41 lots:

Component	SF
Dwellings, garages, porches, rear porches and patios (41 footprints from data/plans . json)	86,448
Lot driveways and entry walks	31,390
Private lane pavement	34,111
Entry drive pavement	4,865
Entrance curb returns	184
Hammerhead turnaround legs	7,200
Guest and mail-kiosk parking bays	2,070

Component	SF
Sidewalks	14,619
Clubhouse roof, court pads, mail kiosk and amenity walks	6,760
less driveway apron / sidewalk overlap counted twice	-3,500
Total	184,148 SF = 4.23 ac = 44.1 % of 417,174 SF

Per lot that is **Plan A 2,752 SF** (1,967 under roof + 72 patio + 653 driveway + 60 walk, 21 lots) and **Plan B 2,898 SF** (2,182 under roof + 656 driveway + 60 walk, 20 lots); driveway areas vary a little with each lot's position on the curving lane. The first 09-03 issue's 188,483 SF / 45.2 % was the same computation at 43 lots and is superseded.

Disturbed area. The ponds were originally sized on an assumed **4.4 disturbed acres**. The real figure, taken off a 2-ft raster of the site less the preserved areas, is **7.541 ac**: raster site area 418,228 SF, less 73,744 SF of 20-ft perimeter buffer bands, less 16,000 SF of creek-woods tract lying outside those bands (the 50-ft undisturbed stream buffer is wholly inside the areas already deducted, so it adds nothing), = 89,744 SF preserved and **328,484 SF disturbed**. That understatement by a factor of about 1.7 is what hid the detention shortfall, and correcting it is what forced the basins to grow — which in turn cost one of the two lots.

Then, with $I = 44.14\%$ ($184,148 \div 417,174$) and $A = 417,174$ SF:

- $R_v = 0.05 + 0.009 \times 44.14 = \mathbf{0.4473}$; $WQ_r = 1.2 \times 0.4473 = \mathbf{0.5367}$ in.
- $WQ_v = \mathbf{0.5367} \div 12 \times 417,174 = \mathbf{18,659}$ cf (whole-site drainage area; the PE will use actual tributary areas, which exclude the undisturbed creek woods and buffers and will lower this). This supersedes the 16,503 cf of 2026-08-29 and the 19,049 cf of the first 09-03 issue.
- **RRv, if the full 1.0-in runoff-reduction credit were earned = $0.4473 \times 1.0 \div 12 \times 417,174 = 15,549$ cf.** The plan as drawn claims **none of it**: layout . json carries $RR_v = 0$ at concept stage, which is the conservative reading and the one this memo uses.
- **Detention required: $10,000$ cf/ac $\times$ 7.541 ac = $75,410$ cf,** with RR_v taken as 0. If the PE earns the full 1.0-in credit the requirement falls to about **59,861 cf**.

D.3 Results, and the basins as now drawn

Item	Value	Inputs
Impervious	44.1 % of the GIS area (against 50–57 % per basin on the 2025 mixed-use hydrology sheet for this same land)	$184,148 \div 417,174$
WQv (treat the 1.2-in runoff)	18,659 cf	$1.2 \times (0.05 + 0.009 \times 44.14) \div 12 \times 417,174$
RRv, if the credit is earned (not claimed on the plan)	15,549 cf	$0.4473 \times 1.0 \div 12 \times 417,174$
Disturbed area	7.541 ac	328,484 SF, D.2
Detention required	75,410 cf	$10,000$ cf/ac $\times$ 7.541 ac, $RR_v = 0$
Detention provided	77,863 cf — Pond 1 33,129 cf + Pond 2 44,734 cf	layout . json ponds
Provided $\div$ required	103.3 % — a margin of 2,453 cf	$77,863 - 75,410$

The basins were enlarged, and that is what cost the second lot. At the 4-ft depth previously drawn the two basins held $22,011 + 22,207 = \mathbf{44,218}$ cf, only 59 % of the requirement; the shortfall was hidden because the sizing assumed 4.4 disturbed acres rather than the 7.541 the plan actually disturbs. Both basins are now **6 ft deep with 3:1 side slopes**, with top of bank **52 $\times$ 180 ft** (Pond 1) and **64 $\times$ 180 ft** (Pond 2) in 20,000-SF tracts, and Pond 2's tract had to grow from 15,000 to 20,000 SF — into ground that had carried a lot. Arithmetic check, prismoidal volume $V = h/6 \cdot (A_{\text{top}} + 4 \cdot A_{\text{mid}} + A_{\text{bottom}})$: for a rectangular 52 $\times$ 180 ft basin, $A_{\text{top}} = 9,360$ SF, $A_{\text{mid}} = 34 \times 162 = 5,508$ SF, $A_{\text{bottom}} = 16 \times 144 = 2,304$ SF $\rightarrow V = 1.0 \times (9,360 + 22,032 + 2,304) = \mathbf{33,696}$ cf; for 64 $\times$ 180 ft, **45,360 cf**. The basins as drawn are trapezoidal rather than rectangular (they follow the tract and the property line), which is why layout . json reports the slightly smaller **33,129** and **44,734 cf**.

What that margin is worth. 3.3 % is not a design margin; it is a screening result on a screening rate. Two things can eat it and one can restore it: rock (the ARE map unit puts gneiss at 22–40 in **under both basins** — a shallow rock line means undercut, blasting, or shallower and longer basins), a stream delineation that moves Pond 2, and, in the other direction, any runoff-reduction credit the PE earns, which is worth up to about 15,500 cf. **Rock probes at both sites before any of this is trusted.** If rock is shallow the fallbacks are a third facility at the terminus green or shallower, longer basins — and either would be preferable to another lot loss.

- **Runoff reduction on clay.** Infiltration will test poorly, so meet RR_v with **bioretention with underdrains** in the pocket greens and lane-side swales, roof-drain disconnection to HOA-maintained yards, and pervious pavers in the guest bays (Table 4.2

permits pervious paving); credit the preserved creek woods and buffers as undisturbed area.

- **Full GCSWMM/GSMM stack for the PE at LDP:** water quality (WQv), runoff reduction (RRv), channel protection (CPv — 24-hour extended detention of the 1-year, 24-hour storm), overbank flood protection (25-year), extreme flood (100-year), downstream analysis to the 10 % point in the Jackson Creek basin, and an erosion, sedimentation and pollution control plan (GSWCC; NPDES GAR100001 — the site exceeds 1 acre and lies within 200 ft of a stream). The 303(d) listing makes **water-quality treatment and no direct discharge into the buffer** the exhibit the neighbours will read.
- **Stream setbacks.** Every lot, pond, pavement edge and structure is screened against **all three digitized stream reaches** — the on-site headwater and both off-site branches — at 75 ft **plus a 5-ft top-of-bank allowance**, i.e. 80 ft from the mapped channel centreline. That allowance exists because the package's distances are measured to a digitized *centreline* while the ordinance measures from *top of bank*; on a 6–16-ft channel the two differ by 3–8 ft. The 2026-08-29 statement that "no lot or pond lies within 75 ft" was true of the on-site reach only and false for one southwest lot against an off-site branch. The screen has been widened to all three reaches and the statement now holds by construction — but it holds because **that lot was removed**: the southwest slot at u 1,230–1,280 is now an unlotted 5,000-SF stream-setback open-space tract carrying no impervious surface, and it is one of the two lots that took the yield from 43 to 41. As drawn, the nearest lot stands 80.6 ft from a mapped channel centreline, the nearest dwelling 106.2 ft, the nearest driveway 102.6 ft, the nearest pond top of bank 80.2 ft and the lane pavement 80.5 ft. **Field delineation still governs** (Memo G).
- **Floodplain:** FEMA Zone X. Table 4.2's prohibition on floodway and floodplain development is met by absence [VERIFY the FIRM panel and effective date at LDP; Lilburn Ch. 109 §109-52 et seq.].
- **Open space is not a stormwater constraint here, and it is not a code requirement either.** The plan provides **141,089 SF = 3.24 ac = 33.8 % of the GIS area** (34.3 % of 9.44 ac) in named HOA tracts that sum exactly to that figure, of which 13,960 SF is built or paved (clubhouse roof, court pads, mail kiosk, amenity walks and the hammerhead legs inside the greens), leaving **127,129 SF = 30.5 %** of green-only open space; the 41,000 SF of buffer easement on lots is additional and is not counted in either figure. Table 4.1's open-space entry points to Development Regulations §5.9, which has now been retrieved: it is titled "**Recreation Areas**" and is triggered only "in single-family detached subdivisions having a **gross area of 50 acres or more** and a minimum lot size less than one acre" (and in duplex subdivisions of 50 acres or more, and attached/townhouse/apartment developments of ten acres or more), requiring "six percent of the gross land area." At **9.44 acres** of single-family detached, §5.9 does not apply. The 33.8 % common open space this plan provides is therefore entirely voluntary — and it rose, not fell, when the two lots came out. **This closes the VERIFY that FACTS §3 carried.** One caution: §5.9's text still refers to "Gwinnett County," "the Board of Commissioners" and "the Department" because Lilburn adopted the county regulations wholesale — confirm with staff who those actors are in a city case [VERIFY].

Appears consistent with the GCSWMM approach at the rezoning level; **all volumes are placeholders until a PE's hydrology (VERIFY the disturbed area, the tributary areas and the rock line).**

Memo E — Fire / Life Safety

Question. What a single 1,754-ft dead-end lane serving 41 one-story dwellings actually triggers under the Georgia fire code, and what to bring to the Gwinnett County Fire Marshal (the City has no fire department).

E.1 The correction that changes this memo

The 2026-08-29 issue stated that the development "exceeds the 30-unit threshold of Appendix D107.1 for a second access road unless the homes are sprinklered," and treated NFPA 13D sprinklers as the required mitigation. **That is the model IFC text, not the rule in force in Georgia.** Ga. Comp. R. & Regs. 120-3-3-.04, "State Minimum Fire Safety Standards with Modifications," Modifications to Chapter 5, item 7:

"Add a new section, Appendix D 107.1, as follows: Developments of one- or two-family dwellings where the number of dwelling units exceeds 120 shall be provided with two separate and approved fire apparatus access roads. Exceptions:
1. Where there are more than 120 dwelling units on a single public or private fire apparatus access road and all dwelling units are equipped throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3, access from two directions shall not be required. ..."

The model IFC 2024 number is 30; **Georgia replaces it with 120.** At **41 dwelling units this development does not trigger the two-access requirement at all**, and no exception needs to be invoked to relieve it. The two lots lost on the corrected site plan move the project further below the threshold, not nearer it. Everything the package previously said about "43 units (> 30) → 13D exception" is withdrawn.

E.2 Inputs

Input	Value	Source
Code	IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04 , Appendices B and D, effective 2026-01-01 [Gwinnett's local adoption of Appendices B and D: VERIFY]	GA DCA <i>Current State Minimum Codes</i> ; SOS rules compilation
Dead-end length	1,754 ft (Arcado Rd right-of-way to the end of pavement, along the travelled way)	layout . json metrics
Units on one access	41 (cap 47 by voluntary condition 2)	same
Lane	22-ft pavement in a 35.9–46.1-ft tract (60 ft through the amenity block), no parking, three hammerhead turnarounds (two 60-ft × 20-ft legs each) at stations 594.5 / 1,164.5 / 1,744.5 ft , so the runs between intersections are 594.5 / 570.0 / 580.0 ft ; entrance radius 25 ft (NE) and 15 ft (SW); design speed 15 mph	layout . json lane, hammerheads, metrics; Sheet C-2.0 governs
Travelled distance to the farthest garage door	≈ 1,701 ft (204-ft entry drive + 1,497 ft of lane from the join at u 149.5 to the last driveway at u ≈ 1,646); end of pavement 1,754 ft	layout . json lots, lane
Buildings	1 story; ridge 18'-6" (Plan A) and 19'-8" (Plan B) above finished grade — 17'-10" and 19'-0" above the finished floor, which is 8 in above grade; clubhouse 1 story, ≈ 2,400 SF	Sheets A-2.1 and A-2.2; docs/09 §5 (data/plans . json still carries a superseded 19'-4" rear-block ridge for Plan B — VERIFY on regeneration)
Sprinklers	NFPA 13D in 100 % of dwellings — voluntary condition 5	data/voluntary-conditions.md

E.3 Code provisions

Provision	Requirement	This project
§503.1.1	fire apparatus access within 150 ft of all exterior walls	every dwelling is within about 100 ft of the lane
§503.2.1 / D102.1	width ≥ 20 ft, vertical clearance 13 ft 6 in, all-weather surface for 75,000 lb	22-ft asphalt lane; no overhead obstructions
Appendix D §D103.4, Table D103.4	0–150 ft: 20 ft wide, no turnaround · 151–500 ft: 20 ft + a 120-ft hammerhead, 60-ft "Y" or 96-ft-diameter cul-de-sac · 501–750 ft: 26 ft + turnaround · Over 750 ft: "Special approval required"	1,754 ft → special approval. The hammerheads match the 120-ft figure; the 22-ft width against the 26 ft the table pairs with long dead ends is the one true relief item
§D103.1	26-ft width where a hydrant is on the access road	hydrants placed at local widenings (Memo C)
§D103.2	grade ≤ 10 % unless approved	steepest existing ground grade on the lane centreline is 10.9 % at u ≈ 1,494 ft, with a second pitch of the same order near u ≈ 1,004; the finished profile must be designed to ≤ 10 % or the grade approved — a PE profile item, not yet designed [VERIFY]
§D103.3	turning radius as specified by the fire code official	25-ft entrance radius on the NE side; hammerhead legs to the D103.1 figure
§D103.6	"NO PARKING — FIRE LANE" signs on 20–26-ft roads	HOA covenant plus signs; 4 spaces per dwelling (164 on lot against the 82 that Table 8.1 requires at 2/DU × 41) and 12 guest and mail-kiosk spaces keep the lane clear
Appendix D §D107.1 (GA)	two access roads where dwelling units exceed 120	41 units — not triggered
§D105 (aerial access, buildings > 30 ft)	not applicable	one story
Gates	none	not gated

E.4 The one relief item, and the routes to it

The only live fire-access question is **special approval of a 1,754-ft dead end at 22 ft of pavement** under Table D103.4. Three routes exist and they are not exclusive:

1. **The Fire Marshal's special approval** under D103.4 itself. There are no published criteria; it is a discretionary decision, which is why it must be obtained in writing before the Planning Commission hearing.
2. **Appendix D107.1 Exception 3 as modified by Ga. Comp. R. & Regs. 120-3-3-.04**, which supplies an explicit menu for "residential developments having less than the minimum of two entrances": enhanced turning radii, **increased road widths**, D103.6 fire-lane signage, an additional **non-vehicular all-weather access road** designed by a registered design professional to Appendix D loading, or **a written statement by the Fire Code Official**.
3. **The statutory waiver, O.C.G.A. §25-2-12(e)(4)**: "the local fire official, building official, or developer may obtain a waiver ... on a waiver form designed and prescribed by the Safety Fire Commissioner. **The State Fire Marshal or designated representative shall respond within 30 days ... If the 30-day time frame is not met, the Plans submitted shall be**

deemed to be approved." The 30-day deemed-approval clock makes this worth filing early rather than late.

E.5 The package to put in front of the Fire Marshal

1. **A 22-ft clear lane with no on-street parking**, signed both sides under D103.6 and enforced by HOA covenant, with **26 ft of pavement offered wherever the strip allows it**. The lane tract widens from 35.9 ft at the entrance to 46.1 ft at the rear, passing 40 ft at about $u = 1,010$; from there north 26 ft of pavement fits alongside both 5-ft sidewalks and their 2-ft strips ($26 + 2 \times 7 = 40$ ft), and it is offered there. Between $u \approx 530$ and $u \approx 1,010$ the tract holds 37 to 40 ft, so 26 ft could be had only by giving up the southwest sidewalk — offered as a trade if the Fire Marshal prefers width to a second walk. In the narrow front reach the tract cannot hold 26 ft plus a 2-ft strip and a 5-ft sidewalk at all, which is precisely the relief being sought.
2. **Three hammerhead turnarounds** at 594.5, 570.0 and 580.0 ft of spacing, the third at the terminus, each two legs of 20 ft $\times$ 60 ft (the Figure D103.1 "120-foot hammerhead").
3. **Hydrants at about 400 ft** on an 8-in main (Memo C), located at the widened points so D103.1's 26-ft width is satisfied where a hydrant stands on the access road.
4. **NFPA 13D sprinklers in every dwelling — offered voluntarily**, and stated as such. They are worth offering for two reasons that survive the D107.1 correction: they halve the required fire flow from 1,000 gpm to 500 gpm on a 1,754-ft dead-end main (Appendix B §B105.1), and they are the strongest single argument for the 22-ft width relief. At roughly \$3,000 per home, 41 homes is about **\$123,000**, and the applicant should say plainly that this is an elective offer and ask the Fire Marshal whether it buys the width. If it does not, the offer should still stand as a life-safety measure on a single-access lane — but the package must not represent it as a code requirement, because it is not.
5. **Clubhouse sprinklered (NFPA 13)** and monitored; Knox box at the clubhouse.
6. **Addressing**: one street name; 4-in reflective numbers visible from the lane; cluster mailboxes at the kiosk.
7. **Grades**: the finished lane profile designed to $\leq 10\%$ (existing ground reaches 10.9 %), 13 ft 6 in vertical clearance, no gate.
8. **Construction sequencing**: the first hammerhead turnaround (station 594.5 ft) built before any certificate of occupancy beyond it, and a temporary turnaround at the Phase-1 limit ($u = 980$ ft). One item belongs on the sequencing sheet and on the demolition plan: **the existing dwelling at 4535 Arcado Rd stands about 1,200 ft back from the right-of-way, at about ($u = 1,202$, $v = -114$) on the ridge — directly on the proposed lane alignment**. It is a Phase-2 demolition. While it stands, the lane physically cannot continue past it, which is a further reason the Phase-1 temporary turnaround at $u = 980$ is not optional. Only 4541 Arcado Rd is near the frontage.
9. **Managed vegetation zone**: keep a 30-ft managed zone behind the rear lots within the buffer-easement rules at the wooded rear [VERIFY with the Fire Marshal; this is not a designated wildland-urban-interface area].

Special approval under D103.4 is a discretionary decision of the fire code official; nothing above may be represented as approval. The plan appears consistent with IFC 2024 (Ga. Comp. R. & Regs. 120-3-3-.04) Appendix D **only if items 1–3 and the D103.4 special approval are accepted in writing** — get the Fire Marshal's letter before the Planning Commission hearing [VERIFY].

Memo F — Schools (Gwinnett County Public Schools)

Question. Student generation of the cottage program against the 2025 mixed-use filing and the as-zoned R-1 baseline.

F.1 Inputs

Input	Value	Source
Cluster	Camp Creek ES / Trickum MS / Parkview HS	FACTS §2 (GIS school zones)
2025 GCPS letter for RZ-2025-01 (104 condominiums), July 2025	+25 ES / +13 MS / +19 HS ; forecasts against capacity 2025-26: Camp Creek 1,414 / 1,325 (+89 over), Trickum 2,203 / 2,275 (–72), Parkview 3,148 / 2,900 (+248 over)	status/rz-2025-01/4817_school-impact-analysis.pdf [VERIFY the figures before citing them]
HOPA 55+	at least 80 % of occupied units with a resident 55+ (24 CFR §100.305(a)), and the fractional unit counts into the 80 % (§100.305(g)): $0.80 \times 41 = 32.8 \rightarrow$ 33 of the 41 units must be occupied by at least one person aged 55 or older ($33 \div 41 = 80.5\%$; 32 would be 78.0 % and would fail). Eight units carry no age requirement. Recorded declaration; published policies; biennial verification (§100.307(c))	42 U.S.C. §3607(b)(2)(C); 24 CFR §§100.305–.307
GCPS per-unit multipliers (single-family detached, not age-restricted)	not in the record	VERIFY by requesting an updated GCPS Planning letter

F.2 Result

- **Proposed (41 HOPA cottages): 0 students** at each of the three schools. A 55+ community with a recorded age-restriction covenant is treated by GCPS as generating no K-12 enrolment; the 2025 letter's +57 total is superseded because the program that produced it — 104 non-age-restricted condominiums — is withdrawn.
- **As-zoned R-1 (37 lots, no age restriction — 4 du/ac × 9.44 ac)** would generate students at GCPS's single-family multipliers, into a cluster where Camp Creek ES and Parkview HS are already forecast over capacity. The cottage program's zero is therefore a net benefit against by-right development, and the §1003-7 "burden on schools" criterion can be answered with a GCPS letter rather than an estimate.
- **Action:** send GCPS Planning the R-2 concept plan and the HOPA condition and ask for a **new impact letter** stating zero students, with the R-1 baseline for comparison, before the Planning Commission hearing.

Memo G — Environmental Screen

Question. Regulated features and the studies the site will need.

G.1 Findings

Feature	Finding	Rule / source	Action
Stream (state waters)	An unnamed order-0 headwater reaches the SW line at u ≈ 1,356–1,392 and ends inside the site at (1,391.7, –209.8); a second branch runs 38–58 ft outside the SW line at u ≈ 868–1,038	Gwinnett GIS hydrology (FACTS §2)	See G.2 — the first question is whether a regulated stream exists here, not only where its bank is
State buffer	25 ft , measured "from the point where vegetation has been wrested by normal stream flow or wave action"; no land-disturbing activity within it	O.C.G.A. §12-7-6(b)(15) ; GA EPD <i>Field Guide for Determining the Presence of State Waters That Require a Buffer</i> (Sept 2017)	No encroachment proposed → no EPD buffer variance needed
City buffer	50-ft undisturbed buffer plus a 25-ft impervious setback (75 ft), measured from the top of the stream bank ; land development in the buffer or impervious cover in the setback "is prohibited unless a variance is granted"; no septic in either	Lilburn Code Ch. 109 Art. VII (MNGWPD model text; section numbers VERIFY — Municode returns HTTP 403 to automated retrieval, transcribe by hand)	Show the 25-ft state line and the 50/75-ft city lines separately on the plan: they are measured from different data
Whether the buffer applies at all	The ordinance begins a "stream" at (1) a spring, seep or groundwater outflow that sustains streamflow, (2) a point with a drainage area of 25 acres or more , or (3) where field studies show a stream in a smaller drainage area. The on-site reach drains far less than 25 acres , so under (2) the regulated stream would begin downstream, off the property; under (1) it begins at a spring or seep that must be located in the field	Lilburn Code Ch. 109 Art. VII / MNGWPD model definitions	This is an open question the package previously treated as settled. The field study must establish first <i>whether and where</i> a stream begins, then the top of bank. The rear lot count and the Pond 2 footprint both turn on the answer
Receiving water	Jackson Creek headwaters (GAR030701030315) — 2024 303(d) Not Supporting (bacteria, biota; urban runoff)	GA EPD 2024 Integrated 305(b)/303(d) list	Emphasise WQv/RRv treatment and no direct discharge into the buffer (Memo D)
Floodplain	FEMA Zone X	Gwinnett GIS [FIRM panel and effective date VERIFY]	Table 4.2 prohibition met by absence; note it on the plan
Wetlands	none mapped in the record (NWI not queried)	—	VERIFY: NWI check plus a wetland delineation at the stream head (hydric indicators are possible despite no hydric map unit)
Soils	AmC2 Appling sandy loam 6–10 % eroded (HSG B); ApB Appling–Hard Labor complex 2–6 %; PfC2 Pacolet-type sandy clay loam eroded; GeB2 / GgE2 Gwinnett/Cecil-type eroded phases (GgE2 at 10–25 % slopes at the rear); ARE Ashlar–Rion–Waterree complex 10–25 % (bedrock 22–40 in) under both pond sites and the mid-swale lane; HYB Helena sandy loam 2–6 % (HSG D, water table ≈ 24 in) at the front SW corner. No hydric units on site.	docs/11 §3.3 (NRCS SDA) [series names VERIFY]	Rock probes at the ponds and the trunk sewer; test pits and an undercut plan at HYB; phased clearing with immediate stabilisation
Trees	Residential subdivisions must file a tree preservation/replacement plan meeting 16 tree density units per acre (exclusive of zoning-buffer acreage); trees ≥ 3 in DBH credited, preserved trees ≥ 12 in earning up to 4× credit; plan by a landscape architect, forester or arborist	Lilburn Code Ch. 109 Art. V §§109-168(3), 109-169, 109-177	Tree survey of the wooded rear and the perimeter buffers; the creek woods, the 20-ft buffers and the terminus green should carry the requirement at multiplied credit

Feature	Finding	Rule / source	Action
Phase I ESA	Two pre-1980 houses with accessory structures. Only 4541 Arcado Rd is near the frontage (footprint centred about u 203, v -174, ≈ 2,396 SF). 4535 Arcado Rd sits about 1,200 ft back from the right-of-way, at about (u 1,202, v -114) on the u 1,080–1,200 ridge, directly on the proposed lane alignment (≈ 4,568 SF footprint) — a Phase-2 demolition standing on the lane , not a frontage structure. Likely septic tanks, possible wells, heating-oil tanks, asbestos and lead paint	ASTM E1527-21; footprints affine-fitted from aerial imagery by tools/existing_conditions.py, approximate, to be located by the RLS survey	Commission a Phase I ESA before closing on 4535/4537; septic abandonment per Gwinnett Environmental Health; asbestos survey before demolition (GA EPD Rule 391-3-14). Sequence the 4535 demolition with the Phase-2 lane construction (Memo E.5 item 8)
Species ; historic	not screened	USFWS IPaC; GA HPD GNAHRGIS	VERIFY: run IPaC and a GNAHRGIS check
Radon	Gwinnett is EPA Radon Zone 1	Georgia Radon Program (2024)	Passive radon stack in every slab (a specification item, not a zoning issue)
Erosion and sediment	site exceeds 1 ac and lies within 200 ft of a stream	GSWCC Manual (2016); NPDES GAR100001	ESPC plan by a certified design professional at LDP; show the 25-ft state buffer on it

G.2 Statement for the application

The site contains no floodplain (FEMA Zone X), no mapped wetlands and no mapped hydric soils. One headwater reach at the rear southwest corner is treated as regulated: the plan preserves it and its 25-ft state buffer, 50-ft undisturbed city buffer and 75-ft impervious setback inside a common "creek woods" tract, and every lot, pond and pavement edge is held **80 ft** from the mapped channel — the 75-ft requirement plus a 5-ft allowance for the top-of-bank location. A field study by the surveyor and the PE, using the GA EPD field guide, will establish whether and where a regulated stream begins and where its top of bank lies; the plan is drawn so that a stream origin anywhere within the mapped reach can be accommodated without moving lots. Subject to that study, a tree survey and a Phase I ESA, the plan appears consistent with Lilburn Code Ch. 109 Art. VII and Art. V.

Summary table (rezoning-level screen)

Memo	Headline	Status
A Traffic	152–203 daily / 8.2–16.0 AM / 12.3–18.3 PM trips at 41–47 units; 34.8 PM as-zoned R-1; ≈ 214 PM for the 2025 MU program	A Level-1 TIS is required (the exemption is 7 lots, not 21 trips); ITE LUC 251 rate UNRESOLVED ; new 24-hour counts required — the 2017 count is ineligible; no left-turn lane (< 75 lots); 251 ft > 244 ft spacing; ISD certificate needed
B Sewer	Phase 1 (24 lots): 0.68 % gravity to the existing on-site MH 927.13 — no extension . Phase 2 (17 lots): only the 478-ft / 178-ft off-site gravity tie at 1.45 % to the Legends main 919.58, ≈ \$168k–\$273k = \$9,900–\$16,000 per Phase-2 lot, up 12 % on the smaller Phase 2	Pumping of every kind foreclosed by DWR policy ; needs a recorded easement, a DWR "Pre-Rezoning" capacity certification and a staff reading of the CPA sewer policy; the drawn alignment is ≈ 20 ft deep where it crosses the lane at u 1,300 — price the low-pocket alternative; invert survey is the first spend
C Water	10,250 gpd average (41 × 250); fire flow 1,000 gpm, or 500 gpm with the voluntary sprinklers ; 6 hydrants; 8-in dead-end main with flushing	DWR flow test
D Storm water	44.1 % impervious (184,148 SF, not 38.4 %); WQv 18,659 cf ; detention 75,410 cf required on 7.541 disturbed acres against 77,863 cf provided — 103.3 % , in two 6-ft basins (33,129 + 44,734 cf); RRv 15,549 cf available but not claimed	Rock probes under both basins; PE hydrology; a 3.3 % margin is a screening result, not a design margin; \$5.9 does not apply — the 33.8 % open space is voluntary
E Fire	1,754-ft dead end → D103.4 special approval ; 22 ft vs 26 ft is the relief item; D107.1 is 120 units in Georgia, so two access roads are not required at 41	Fire Marshal's letter; sprinklers are a voluntary offer (41 × ≈ \$3,000 ≈ \$123,000), not a required mitigation; 4535 Arcado Rd stands on the lane alignment and is a Phase-2 demolition
F Schools	0 students (HOPA; 33 of the 41 units must house a 55+ resident — 0.80 × 41 = 32.8, rounded into the 80 %)	New GCPS letter
G Environmental	25/50/75-ft stream buffers preserved with an 80-ft design screen; Zone X; no hydric soils; 16 TDU/ac tree plan; Phase I ESA	Field study must establish whether a regulated stream begins on site

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.